Document verification accuracy regression

Report 0dfd35fa-06f5-4f28-b68a-df142b9c643b
Generated 2026-10-17T03:29:22.513939+00:00
Outcome: hypothesis_report

Working explanation

The recorded change in verifier/preprocessing.py from [thumbnail = resize(image, (224, 224), interpolation=BICUBIC)] to [thumbnail = resize(image, (224, 224), interpolation=NEAREST)] is the leading candidate explanation for the reported issue; causality still requires controlled reproduction.

Verified observations

VERIFIED / high confidence

Recorded accuracy changed from 98.00% (980/1000) in v2.2 to 91.00% (910/1000) in v2.3; change -7.00 percentage points.

Supporting sources: [6]
Contradictory/context sources: None

VERIFIED / high confidence

The high_resolution subgroup changed by 0.00 percentage points between v2.2 and v2.3.

Supporting sources: [7]
Contradictory/context sources: None

VERIFIED / high confidence

The low_resolution subgroup changed by -23.33 percentage points between v2.2 and v2.3.

Supporting sources: [7]
Contradictory/context sources: None

VERIFIED / high confidence

The deployment configuration fields {"high_resolution_fraction": 0.7, "low_resolution_fraction": 0.3, "model_hash": "weights-4b17ac", "threshold": 0.85} were unchanged between v2.2 and v2.3.

Supporting sources: [9]
Contradictory/context sources: None

Root-cause hypotheses

HYPOTHESIS / high confidence

The recorded change in verifier/preprocessing.py from [thumbnail = resize(image, (224, 224), interpolation=BICUBIC)] to [thumbnail = resize(image, (224, 224), interpolation=NEAREST)] is the leading candidate explanation for the reported issue; causality still requires controlled reproduction.

Supporting sources: [10], [6]
Contradictory/context sources: None

Recommended actions

1. Reconsider preprocessing changes: The switch from bicubic to nearest interpolation in `verifier/preprocessing.py` may have affected low-resolution image quality, leading to lower verification accuracy.

2. Check model weights: The model hash remains `weights-4b17ac`, so no change in model weights is evident.

3. Monitor low-resolution subgroup: The significant drop in the low-resolution subgroup (-23.33%) suggests a problem with preprocessing or model behavior in low-resolution scenarios.

Limitations and uncertainty

1. This investigation compared the selected recorded sources; no controlled causal experiment was performed.

2. The model selected among observed changes. Unrecorded causes may remain; qualitative confidence is not a calibrated probability.

3. Repository evidence comes from explicitly labeled synthetic source records.

4. Citation identity was checked programmatically; semantic entailment and causal correctness still require human review and controlled reproduction.

5. Model-authored explanations are hypotheses. No corrective action has been executed by this investigation.

Source evidence

Citation numbers below refer to recorded source outputs. Citation presence does not establish semantic entailment or causality.

[1] monitoring-runbook.md

ID: 74740724-73a7-531c-9564-57bf82a74cc6
Source: knowledge

{"chunk_id": "74740724-73a7-531c-9564-57bf82a74cc6", "document_id": "2d85ea2a-39cd-47e8-b001-64b9535305b9", "filename": "monitoring-runbook.md", "graph_expanded": false, "page": 0, "section": "Document verification monitoring runbook", "sha256": "899d0497136953e0b81fd936403565d7c35d67ed34ee6481dbfa6fb3b8cbaf3a"}

```
# Document verification monitoring runbook

Synthetic operations document. Service: document-verification. Deployment: v2.3.

When accuracy changes, compare releases on the same sample strata. Inspect
deployment configuration, preprocessing changes, warning distributions, latency,
model hashes and decision thresholds. Input drift and code regressions are
competing hypotheses, not interchangeable explanations.

The stroke_preservation warning measures lost high-frequency image structure in
low-resolution scans. A warning does not independently prove a causal regression.
Re-run affected samples with both preprocessing configurations to validate a fix.
```

```
Write actions, including issue creation or rollback, require explicit approval.
```

[2] release-v2.3.md

ID: 977bca57-18ba-5bba-84b1-e29ebe2e765b
Source: knowledge

{"chunk_id": "977bca57-18ba-5bba-84b1-e29ebe2e765b", "document_id": "7ed8f124-272c-4da2-a6d5-d0efe346ae41", "filename": "release-v2.3.md", "graph_expanded": false, "page": 0, "section": "Release v2.3 rollout notes", "sha256": "6f49755e5741f9a46eaf045039f3fcb7307614f1d17273c54284c941046053f7"}

```
# Release v2.3 rollout notes

Synthetic deployment record. Service: document-verification. Deployment: v2.3.
Repository: demo/document-verification. Model: verifier-2026-09.

Release v2.3 includes request tracing, a thumbnail-path optimization, updated
dashboard labels and a dependency lock refresh. The model checkpoint remains
verifier-2026-09 with SHA256 weights-4b17ac. Decision threshold remains 0.85.
Document traffic still contains 70 percent high-resolution and 30 percent
low-resolution scans. No model retraining or new training corpus was deployed.

The rollout completed on 2026-10-10 at 10:00 UTC. Compare subgroup metrics and
image preprocessing parity before deciding whether the model itself changed.
```

[3] preprocessing-contract.md

ID: 70ce8b5b-4236-553a-a911-8ae4252ecd03
Source: knowledge

{"chunk_id": "70ce8b5b-4236-553a-a911-8ae4252ecd03", "document_id": "df9b2673-9b21-4cb2-b1e1-9a8ec120ee2b", "filename": "preprocessing-contract.md", "graph_expanded": false, "page": 0, "section": "Document preprocessing contract", "sha256": "a8d1917418d40e2d725b1338d607ac5e01dcc7ac471d0cec412717f5f22596dd"}

```
# Document preprocessing contract

Synthetic reference document. Service: document-verification. Model: verifier-2026-09.
Repository: demo/document-verification. Deployment: v2.2.

The verifier accepts 224 by 224 RGB tensors. Production document thumbnails must
use bicubic interpolation before normalization. Training and evaluation both used
bicubic interpolation. Fine character strokes in low-resolution scans are sensitive
to resampling. Nearest-neighbor is intended only for segmentation masks, not RGB
document thumbnails. Normalization mean/std and decision threshold are versioned.

Validate preprocessing parity, stratified accuracy and source-image quality before
approving a release. A global score can hide subgroup-specific failures.
```

[4] traffic-summary.md

ID: 580d92ac-7c5d-536a-9780-824c2c0c9539
Source: knowledge

{"chunk_id": "580d92ac-7c5d-536a-9780-824c2c0c9539", "document_id": "87fd4f37-ae25-40c4-96c1-073f0d35f106", "filename": "traffic-summary.md", "graph_expanded": false, "page": 0, "section": "Incoming document population", "sha256": "9789c9252766117cabb669c6e236d69bb62dc2bf653882dee97d748cead28737"}

```
# Incoming document population

Synthetic source-quality summary. Service: document-verification.

The v2.2 evaluation window and v2.3 evaluation window each contain 1,000 labeled
documents. Each window has 700 high-resolution and 300 low-resolution scans.
Country mix, language mix and mean source resolution are stable by stratum.
Labels were produced by the same synthetic labeling policy in both windows.

This comparison rules against a large change in category mix, but cannot exclude
every unmeasured distribution shift. Review subgroup errors and changed code.
```

[5] unrelated-maintenance.md

ID: f20bd4df-2aef-5d35-bc0c-a4d24c63db2c
Source: knowledge

{"chunk_id": "f20bd4df-2aef-5d35-bc0c-a4d24c63db2c", "document_id": "7d615610-75c7-4ce0-9c7d-f3c4d95f0172", "filename": "unrelated-maintenance.md", "graph_expanded": false, "page": 0, "section": "Cafeteria menu deployment", "sha256": "aaa89445376b29b35325b22efc6a181860db0f3cb7cbe505337bb9735645479a"}

```
# Cafeteria menu deployment

Synthetic irrelevant document for retrieval evaluation. Service: cafeteria-menu.
Deployment: menu-2026-10. Repository: demo/menu-service.

The lunch menu added a vegetarian option. This service does not process document
images, share model weights or participate in the document verification pipeline.
```

[6] get model metrics

ID: get_model_metrics:83221eab6120659910f9bb82
Source: get_model_metrics

{"query": {"version": null}, "tool": "get_model_metrics", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}


```
{"metrics": [{"version": "v2.2", "samples": 1000, "correct": 980, "accuracy": 0.98, "confidence":
0.9331294999999998, "latency_ms": 52.41593999999994}, {"version": "v2.3", "samples": 1000, "correct":
910, "accuracy": 0.91, "confidence": 0.9164059999999998, "latency_ms": 42.485379999999985}]}
```


[7] compare model versions

ID: compare_model_versions:40cb5ce5bf142898906b0512
Source: compare_model_versions

{"query": {"baseline": "v2.2", "candidate": "v2.3"}, "tool": "compare_model_versions", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}


```
{"baseline": "v2.2", "candidate": "v2.3", "categories": [{"category": "high_resolution", "baseline":
{"version": "v2.2", "category": "high_resolution", "samples": 700, "correct": 686, "accuracy": 0.98},
"candidate": {"version": "v2.3", "category": "high_resolution", "samples": 700, "correct": 686,
"accuracy": 0.98}, "change_percentage_points": 0.0, "approximate_95_percent_interval_pp": [-1.467,
1.467]}, {"category": "low_resolution", "baseline": {"version": "v2.2", "category": "low_resolution",
"samples": 300, "correct": 294, "accuracy": 0.98}, "candidate": {"version": "v2.3", "category":
"low_resolution", "samples": 300, "correct": 224, "accuracy": 0.7466666666666667},
"change_percentage_points": -23.333, "approximate_95_percent_interval_pp": [-28.504, -18.163]}],
"limitation": "Normal-approximation intervals describe association, not causation; small samples and
dependent observations require additional analysis."}
```

[8] get failure distribution

ID: get_failure_distribution:e17e293854a56971eb0ffde0
Source: get_failure_distribution

{"query": {"version": "v2.3"}, "tool": "get_failure_distribution", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}


```
{"categories": [{"category": "low_resolution", "samples": 300, "failures": 76}, {"category":
"high_resolution", "samples": 700, "failures": 14}]}
```


[9] get deployment statistics

ID: get_deployment_statistics:a164b4d758b02211fe3d3759
Source: get_deployment_statistics

{"query": {}, "tool": "get_deployment_statistics", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}


```
{"deployments": [{"id": "b68fb6bf-f6e0-4a50-9a14-2a4bd203c177", "version": "v2.3", "service":
"document-verification", "commit_sha": "bbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb", "deployed_at":
"2026-10-10T10:00:00+00:00", "configuration": {"model_hash": "weights-4b17ac", "threshold": 0.85,
"high_resolution_fraction": 0.7, "low_resolution_fraction": 0.3, "synthetic": true}}, {"id":
"cd109f40-2b4b-403e-a434-b4bbab883d34", "version": "v2.2", "service": "document-verification",
"commit_sha": "aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa", "deployed_at": "2026-10-03T10:00:00+00:00",
"configuration": {"model_hash": "weights-4b17ac", "threshold": 0.85, "high_resolution_fraction": 0.7,
"low_resolution_fraction": 0.3, "synthetic": true}}]}
```


[10] Recorded repository changes

ID: get_recent_commits:c314461a3e513c06b93e355e
Source: get_recent_commits

{"mode": "synthetic", "repository": "demo/document-verification"}


```
{"commits": [{"repository": "demo/document-verification", "sha":
"bbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb", "message": "Optimize thumbnail preprocessing path",
"files": ["verifier/preprocessing.py"], "diff": "--- verifier/preprocessing.py\n+++
verifier/preprocessing.py\n- thumbnail = resize(image, (224, 224), interpolation=BICUBIC)\n+
thumbnail = resize(image, (224, 224), interpolation=NEAREST)\n tensor = normalize(thumbnail,
TRAINING_MEAN, TRAINING_STD)", "committed_at": "2026-10-10T07:00:00+00:00", "id":
"18d475f6-6d37-4191-a335-c4bff153ef88", "created_at": "2026-10-17T03:26:46.914248+00:00",
"workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"repository": "demo/document-verification",
"sha": "ccccccccccccccccccccccccccccccccccccccc", "message": "Add structured request tracing",
"files": ["verifier/telemetry.py"], "diff": "+ span.set_attribute('deployment.version', version)",
"committed_at": "2026-10-10T06:00:00+00:00", "id": "abc98a6d-c6c8-4fbf-b918-fa34161561b0",
"created_at": "2026-10-17T03:26:46.914257+00:00", "workspace_id":
"cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"repository": "demo/document-verification", "sha":
"ddddddddddddddddddddddddddddddddddddddd", "message": "Clarify dashboard labels", "files":
["frontend/labels.ts"], "diff": "- label: 'Verification score'\n+ label: 'Verification confidence'",
"committed_at": "2026-10-10T05:00:00+00:00", "id": "69905622-8430-4e7b-a591-1f5b9818946c",
"created_at": "2026-10-17T03:26:46.914261+00:00", "workspace_id":
"cb0c571f-d64a-4047-b69c-9de6b20c28ca"}]}
```


[11] get application logs

ID: get_application_logs:df605db99af585a68c42e81e
Source: get_application_logs

{"tool": "get_application_logs"}

{"logs": [{"service": "document-verification", "version": "v2.3", "level": "warning", "message": "stroke_preservation below 0.30 for low-resolution thumbnail", "observed_at": "2026-10-10T11:39:00+00:00", "metadata_json": {"category": "low_resolution", "synthetic": true}, "id": "703d81dd-cc2f-4e38-834d-3f554d01d142", "created_at": "2026-10-17T03:26:46.511168+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "warning", "message": "stroke_preservation below 0.30 for low-resolution thumbnail", "observed_at": "2026-10-10T11:38:00+00:00", "metadata_json": {"category": "low_resolution", "synthetic": true}, "id": "a46790b3-8b5b-42f5-82ff-559460b9fb36", "created_at": "2026-10-17T03:26:46.511163+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "warning", "message": "stroke_preservation below 0.30 for low-resolution thumbnail", "observed_at": "2026-10-10T11:37:00+00:00", "metadata_json": {"category": "low_resolution", "synthetic": true}, "id": "af27f8ec-a12f-4d5c-b4cb-7cb070035b2f", "created_at": "2026-10-17T03:26:46.511160+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "info", "message": "verification request completed", "observed_at": "2026-10-10T11:36:00+00:00", "metadata_json": {"category": "high_resolution", "synthetic": true}, "id": "5e2d20a5-9baf-43ba-b504-338fa64cf2a9", "created_at": "2026-10-17T03:26:46.511156+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "warning", "message": "stroke_preservation below 0.30 for low-resolution thumbnail", "observed_at": "2026-10-10T11:35:00+00:00", "metadata_json": {"category": "low_resolution", "synthetic": true}, "id": "3065b511-1493-41cd-9998-badaba331a49", "created_at": "2026-10-17T03:26:46.511152+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "warning", "message": "stroke_preservation below 0.30 for low-resolution thumbnail", "observed_at": "2026-10-10T11:34:00+00:00", "metadata_json": {"category": "low_resolution", "synthetic": true}, "id": "545b48be-81bf-4550-a6ba-2400ad3c8a8e", "created_at": "2026-10-17T03:26:46.511148+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "warning", "message": "stroke_preservation below 0.30 for low-resolution thumbnail", "observed_at": "2026-10-10T11:33:00+00:00", "metadata_json": {"category": "low_resolution", "synthetic": true}, "id": "dc7a6605-7f84-4570-94a6-b393fb3c7a37", "created_at": "2026-10-17T03:26:46.511144+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "info", "message": "verification request completed", "observed_at": "2026-10-10T11:32:00+00:00", "metadata_json": {"category": "high_resolution", "synthetic": true}, "id": "88c84792-9073-4a3e-9eea-647a60998b20", "created_at": "2026-10-17T03:26:46.511140+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "warning", "message": "stroke_preservation below 0.30 for low-resolution thumbnail", "observed_at": "2026-10-10T11:31:00+00:00", "metadata_json": {"category": "low_resolution", "synthetic": true}, "id": "f31049de-c81c-44fe-9da8-1fbdf75f431f", "created_at": "2026-10-17T03:26:46.511136+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "warning", "message": "stroke_preservation below 0.30 for low-resolution thumbnail", "observed_at": "2026-10-10T11:30:00+00:00", "metadata_json": {"category": "low_resolution", "synthetic": true}, "id": "f6f95b25-5629-4d0a-9a61-8a746d66d520", "created_

at": "2026-10-17T03:26:46.511132+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "warning", "message": "stroke_preservation below 0.30 for low-resolution thumbnail", "observed_at": "2026-10-10T11:29:00+00:00", "metadata_json": {"category": "low_resolution", "synthetic": true}, "id": "5a52b4b8-a76d-46a4-b2a4-521f0303c606", "created_at": "2026-10-17T03:26:46.511129+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "info", "message": "verification request completed", "observed_at": "2026-10-10T11:28:00+00:00", "metadata_json": {"category": "high_resolution", "synthetic": true}, "id": "8a25eb26-b58b-4718-b1b0-c0149979d0bb", "created_at": "2026-10-17T03:26:46.511125+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "warning", "message": "stroke_preservation below 0.30 for low-resolution thumbnail", "observed_at": "2026-10-10T11:27:00+00:00", "metadata_json": {"category": "low_resolution", "synthetic": true}, "id": "ca8eca8d-d9e9-468d-86eb-5cbfb370b21d", "created_at": "2026-10-17T03:26:46.511120+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "warning", "message":

"stroke_preservation below 0.30 for low-resolution thumbnail", "observed_at": "2026-10-10T11:26:00+00:00", "metadata_json": {"category": "low_resolution", "synthetic": true}, "id": "f2c04ddb-6501-4e19-b114-0c7017160127", "created_at": "2026-10-17T03:26:46.511117+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "warning", "message": "stroke_preservation below 0.30 for low-resolution thumbnail", "observed_at": "2026-10-10T11:25:00+00:00", "metadata_json": {"category": "low_resolution", "synthetic": true}, "id": "0a2b2a50-3205-452c-92f1-45bb80a99b4e", "created_at": "2026-10-17T03:26:46.511113+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "info", "message": "verification request completed", "observed_at": "2026-10-10T11:24:00+00:00", "metadata_json": {"category": "high_resolution", "synthetic": true}, "id": "d49b65e4-87a6-45e6-ae6e-6890f56112e3", "created_at": "2026-10-17T03:26:46.511109+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "warning", "message": "stroke_preservation below 0.30 for low-resolution thumbnail", "observed_at": "2026-10-10T11:23:00+00:00", "metadata_json": {"category": "low_resolution", "synthetic": true}, "id": "7b4b1c48-08c7-40ae-b1be-b1ef2a3490fc", "created_at": "2026-10-17T03:26:46.511105+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "warning", "message": "stroke_preservation below 0.30 for low-resolution thumbnail", "observed_at": "2026-10-10T11:22:00+00:00", "metadata_json": {"category": "low_resolution", "synthetic": true}, "id": "24e83c3d-9fa8-4e8a-8931-d1505005c3b3", "created_at": "2026-10-17T03:26:46.511101+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "warning", "message": "stroke_preservation below 0.30 for low-resolution thumbnail", "observed_at": "2026-10-10T11:21:00+00:00", "metadata_json": {"category": "low_resolution", "synthetic": true}, "id": "f1300cf1-42f4-4bcf-804e-a374afff81b1", "created_at": "2026-10-17T03:26:46.511096+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}, {"service": "document-verification", "version": "v2.3", "level": "info", "message": "verification request completed", "observed_at": "2026-10-10T11:20:00+00:00", "metadata_json": {"category": "high_resolution", "synthetic": true}

, "id": "706d858b-8593-4d1a-a56f-cb3e3fcb9781", "created_at": "2026-10-17T03:26:46.511093+00:00", "workspace_id": "cb0c571f-d64a-4047-b69c-9de6b20c28ca"}]}

[12] get error statistics

ID: get_error_statistics:d3de6ea82f592e829d5f1876
Source: get_error_statistics

{"tool": "get_error_statistics"}

{"counts": {"v2.3:warning": 23, "v2.3:info": 7, "v2.2:info": 10}, "sample_size": 40}